

Solution Requirements

Date: 30 September 2026 | Team ID: 6ab2240ffc67a5bc0222524ef | Project: ComicCraft – AI Comic Story Creator

Functional requirements

S.No	Requirement category	Requirement description	Priority
1	Input validation	Accept story prompt, character name, setting, tone and art style with Pydantic validation.	High
2	Comic outline	Create exactly five panel outline records with title, scene description and image prompt.	High
3	Story generation	Create caption, narration and dialogue for each panel.	High
4	Image generation	Generate one illustration per panel and save it under static/panels.	High
5	Layout	Match outline, story and image assets into a five-panel layout.	High
6	Preview	Render the generated comic in the web interface.	High
7	PDF export	Compile panel image and text content into a multi-page PDF.	High
8	JSON API	Support /generate-comic/json for API clients.	Medium
9	Image testing	Provide /test-image for direct image-generation testing.	Medium
10	Error handling	Return useful HTTP errors for invalid input or generation failures.	High

Non-functional requirements

Area	Requirement
Usability	The form should be understandable to a first-time creator and the preview should be readable.
Maintainability	Generation, layout, export, routing and validation are separated into modules.
Reliability	The application validates inputs and checks that exactly five panels are assembled.
Security	API credentials should be supplied through environment variables rather than committed to source.
Performance	Image generation should be isolated as a separate stage because it can dominate total response time.
Portability	The project can be run with Python, a virtual environment and the dependencies in requirements.txt.